

EMMETT COLLINGS

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EDUCATION

University of Calgary

September 2019 – December 2022

Bachelor of Science, Major in Computer Science

Calgary, AB

- GPA: 3.8/4.0, focus in theoretical computer science
- Elective Highlights: Computer Security, Distributed Algorithms, Randomized Algorithms, Compiler Construction, Complexity Theory, Network Analysis, Retrogames

EXPERIENCE

Software Developer - Safety Critical Systems

May 2023 – June 2025

Hexagon A&P

Calgary, AB

- Developed GNSS corrections integrity monitoring application in C++ for use in autonomous driving technologies
 - * Contributed to all stages of project, including requirements gathering, architectural and unit design, implementation, and testing
- Participated in internal and external audits to ensure compliance with ASPICE and ISO 26262 safety standards
- Led improvement and creation of in house development tools aimed at accelerating development, improving code quality, and adhering to industry safety standards
- Technologies used: C++, CMake, Python, Cantata, Perforce

C++ Software Developer

May 2022 – August 2022

Nobal Technologies

Calgary, AB

- Worked with hardware and software to create and maintain smart mirror devices and their applications
- Large role in creation of smart mirror hospitality application
 - * Contributed to system design for major rewrite of existing codebase for previous applications
 - * Hospitality app provides functionality such as check in/out, ride share calling, room service, ticket booking
 - * Integrated with several third party software platforms to provide the above functionality
- Technologies used: C++, Qt, QML

Software Developer Intern

May 2021 – August 2021

Neo Financial

Calgary, AB

- Worked with developer experience (DX) to create internal tools to aid developer efficiency
- Created private code coverage tool (github action + web app) that tracked test code coverage of repositories
- Created developer contribution tracking web app, which kept leaderboards of code reviews, PRs merged, etc.
- Technologies used: Typescript, Node.js, MongoDB, Next.js

Private Tutor

September 2017 – June 2018

Self Employed

Calgary, AB

- Specialized in high school physics and mathematics

TECHNICAL SKILLS

Languages: C++, C, Python, Assembly, Typescript/Javascript

Developer Tools: git, Perforce, CMake, Jenkins, Artifactory, Linux, Windows

Other Training: ASPICE, MISRA C++, GNSS Overview

RESEARCH & PROJECTS

Bubbie-VI | 6502 Assembly

September 2022 – December 2022

- A game written in 6502 assembly for the Commodore VIC-20
- Players must retrieve lost pet ducks in a dungeon while dodging enemies
- Resource constrained (3.5K of memory available on the VIC-20)
 - * Created ad hoc data compression schemes
 - * Required efficient and small code
 - * Uses self modification on code and data to increase space efficiency further

NHL Linemates Network | Python, Jupyter

September 2022 – December 2022

- A network analysis of the effects that hockey players have on their linemates

- The links between players are investigated using network science techniques such as community detection and degree centrality to determine players that effectively improve their teammates
- Makes use of data and network science as well as statistics for analysis
- Results summarized in an academic paper which can be found on my GitHub

Web Proxy | *C, socket programming*

January 2022

- Intercepts web traffic and replaced all jpgs from incoming HTTP responses to pictures of clowns
- Swapped plaintext to various joke phrases as well

INTERESTS & AWARDS

Interests

- Reading – Classics, non-fiction
- Sports – rock climbing, volleyball, badminton, swimming
- Outdoor – rock climbing, backpacking, scrambling

Awards

- Hexagon A&P High Five - Given to recognize employees internally (2025)
- Deans List (2018, W2020, W2021)
- National AP Scholar (2015)
- National Swimming Championship Qualifier (2013-16)
- High School Div 3 City Champion – Volleyball (2014)